Stoke's Theorem

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Curl

- Note: Gauss law-differential volume element – Divergence
- Here, Amperes law - differential surface element – Curl

$$\oint \mathbf{H} \cdot d\mathbf{L} = I_{encl}$$

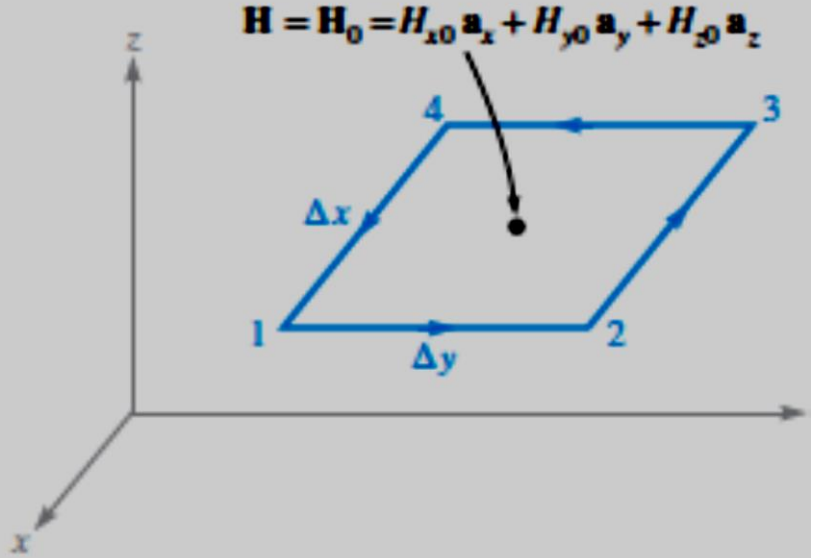


Figure 7.13 An incremental closed path in rectangular coordinates is selected for the application of Ampère's circuital law to determine the spatial rate of change of **H**.

H at the *center* of this small rectangle

$$\mathbf{H}_0 = H_{x0}\mathbf{a}_x + H_{y0}\mathbf{a}_y + H_{z0}\mathbf{a}_z$$

The closed line integral of **H** about this path is then approximately the sum of the four values of $\mathbf{H} \cdot \Delta\mathbf{L}$ on each side.

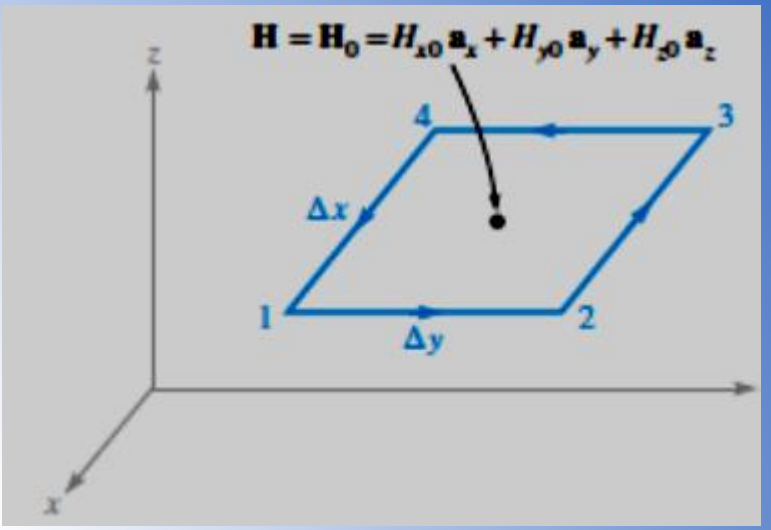
We choose the direction of traverse as 1-2-3-4-1, which corresponds to a current in the $\mathbf{a}_z$ direction, and the first contribution is therefore

$$(\mathbf{H} \cdot \Delta\mathbf{L})_{1-2} = H_{y,1-2}\Delta y$$

$$(\mathbf{H}\cdot\Delta\mathbf{L})_{1-2}=H_{y,1-2}\Delta y$$

$$H_{y,1-2}\doteq H_{y0}+\frac{\partial H_y}{\partial x}\left(\frac{1}{2}\Delta x\right)$$

$$(H\cdot\Delta L)_{1-2}\doteq\left(H_{y0}+\frac{1}{2}\frac{\partial H_y}{\partial x}\Delta x\right)\Delta y$$



Similarly $$(\mathbf{H}\cdot\Delta\mathbf{L})_{2-3}\doteq H_{x,2-3}(-\Delta x)\doteq-\left(H_{x0}+\frac{1}{2}\frac{\partial H_x}{\partial y}\Delta y\right)\Delta x$$

Continuing for the remaining two segments and adding the results,

$$\oint\mathbf{H}\cdot d\mathbf{L}\doteq\left(\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\right)\Delta x\Delta y$$

current density $\mathbf{J}$, the enclosed current is then $\Delta I\doteq J_z\Delta x\Delta y$, and

$$\oint\mathbf{H}\cdot d\mathbf{L}\doteq\left(\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\right)\Delta x\Delta y\doteq J_z\Delta x\Delta y$$

or

$$\frac{\oint\mathbf{H}\cdot d\mathbf{L}}{\Delta x\Delta y}\doteq\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\doteq J_z$$

- As we cause the closed path to shrink

$$\lim_{\Delta x,\Delta y\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta x\Delta y}=\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}=J_z$$

- Analogous processes lead to expressions for the *x* and *y* components of the current density,

$$\lim_{\Delta y,\Delta z\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta y\Delta z}=\frac{\partial H_z}{\partial y}-\frac{\partial H_y}{\partial z}=J_x$$

$$\lim_{\Delta z,\Delta x\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta z\Delta x}=\frac{\partial H_x}{\partial z}-\frac{\partial H_z}{\partial x}=J_y$$

- Given

$$(\text{curl }\mathbf{H})_N=\lim_{\Delta S_N\to 0}\frac{\oint \mathbf{H}\cdot d\mathbf{L}}{\Delta S_N}$$

$$\text{curl }\mathbf{H}=\left(\frac{\partial H_z}{\partial y}-\frac{\partial H_y}{\partial z}\right)\mathbf{a}_x+\left(\frac{\partial H_x}{\partial z}-\frac{\partial H_z}{\partial x}\right)\mathbf{a}_y+\left(\frac{\partial H_y}{\partial x}-\frac{\partial H_x}{\partial y}\right)\mathbf{a}_z$$

$$\text{curl }\mathbf{H}=\begin{vmatrix}\mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z\\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z}\\ H_x & H_y & H_z\end{vmatrix}$$

vector operator

$$\text{curl }\mathbf{H}=\nabla\times\mathbf{H}$$

Curl in cylindrical and spherical coordinates

$$\nabla \times \mathbf{H} = \left(\frac{1}{\rho}\frac{\partial H_z}{\partial \phi} - \frac{\partial H_\phi}{\partial z}\right)\mathbf{a}_\rho + \left(\frac{\partial H_\rho}{\partial z} - \frac{\partial H_z}{\partial \rho}\right)\mathbf{a}_\phi + \left(\frac{1}{\rho}\frac{\partial (\rho H_\phi)}{\partial \rho} - \frac{1}{\rho}\frac{\partial H_\rho}{\partial \phi}\right)\mathbf{a}_z \quad \text{(cylindrical)}$$

$$\nabla \times \mathbf{H} = \frac{1}{r\sin\theta}\left(\frac{\partial (H_\phi \sin\theta)}{\partial \theta} - \frac{\partial H_\theta}{\partial \phi}\right)\mathbf{a}_r + \frac{1}{r}\left(\frac{1}{\sin\theta}\frac{\partial H_r}{\partial \phi} - \frac{\partial (rH_\phi)}{\partial r}\right)\mathbf{a}_\theta + \frac{1}{r}\left(\frac{\partial (rH_\theta)}{\partial r} - \frac{\partial H_r}{\partial \theta}\right)\mathbf{a}_\phi \quad \text{(spherical)}$$

Curl Formula in Cylindrical

$$\nabla \times \vec{A} = \frac{1}{\rho}\begin{vmatrix} \hat{a}_\rho & \rho\hat{a}_\phi & \hat{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

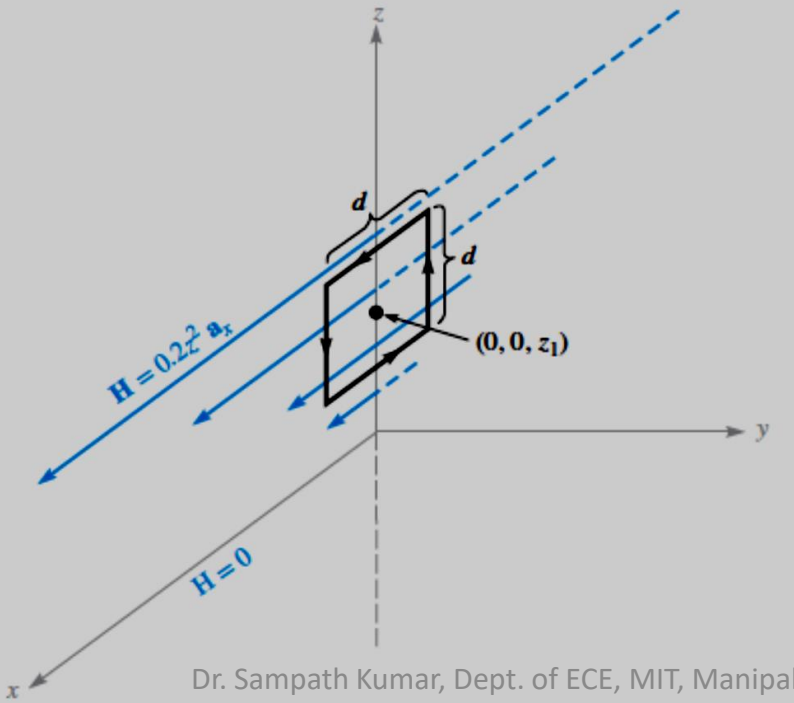
Curl Formula in Spherical

$$\nabla \times \vec{A} = \frac{1}{r^2 \sin\theta}\begin{vmatrix} \hat{a}_r & r\hat{a}_\theta & r\sin\theta\hat{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & rA_\theta & r\sin\theta A_\phi \end{vmatrix}$$

Try this solved example from the textbook

EXAMPLE 7.2

As an example of the evaluation of curl **H** from the definition and of the evaluation of another line integral, suppose that $\mathbf{H} = 0.2z^2\mathbf{a}_x$ for $z > 0$, and $\mathbf{H} = 0$ elsewhere, as shown in Figure 7.15. Calculate $\oint \mathbf{H} \cdot d\mathbf{L}$ about a square path with side d, centered at $(0, 0, z_1)$ in the $y = 0$ plane where $z_1 > d/2$.



$$\nabla \times \mathbf{H} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 0.2z^2 & 0 & 0 \end{vmatrix} = \frac{\partial}{\partial z}(0.2z^2)\mathbf{a}_y = 0.4z\mathbf{a}_y$$

Example 3.14

The magnetic field intensity in free space is given by $\mathbf{H} = 10\rho^2\,\mathbf{a}_\phi$ (A/m).

a. Find **J**.

b. Integrate **J** over the circular surface, $\rho = 1$, all ϕ, and $z = 0$.

point form of Ampère's circuital law,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

$$\mathbf{J} = \nabla \times \mathbf{H}$$

$$= \frac{1}{\rho}\begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ H_\rho & \rho H_\phi & H_z \end{vmatrix}$$

$$= \frac{1}{\rho}\begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial\rho} & 0 & 0 \\ 0 & \rho H_\phi & 0 \end{vmatrix}$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}[\rho H_\phi]\mathbf{a}_z$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}[\rho \times 10\rho^2]\,\mathbf{a}_z$$

$$= 30\rho\,\mathbf{a}_z \ (\mathrm{A/m^2})$$

$$I = \int\int \mathbf{J}\cdot d\mathbf{S}$$

$$= \int_{\rho=0}^{1}\int_{\phi=0}^{2\pi} 30\rho\,\mathbf{a}_z \cdot \rho d\rho d\phi\,\mathbf{a}_z$$

$$= 30 \times 2\pi \int_{\rho=0}^{1} \rho^2\,d\rho$$

$$= 30 \times 2\pi \times \frac{1}{3} = 20\pi \ (\mathrm{A}).$$

point form of Ampère's circuital law,

$$\nabla \times \mathbf{H} = \mathbf{J}$$

STOKES' THEOREM:

$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

$$\oint_S \mathbf{D} \cdot d\mathbf{S} = \int_{\text{vol}} \nabla \cdot \mathbf{D}\, dv$$

Note: Similar to Divergence theorem which relates volume integral and surface integral.
Here surface integral and line integrals are related.

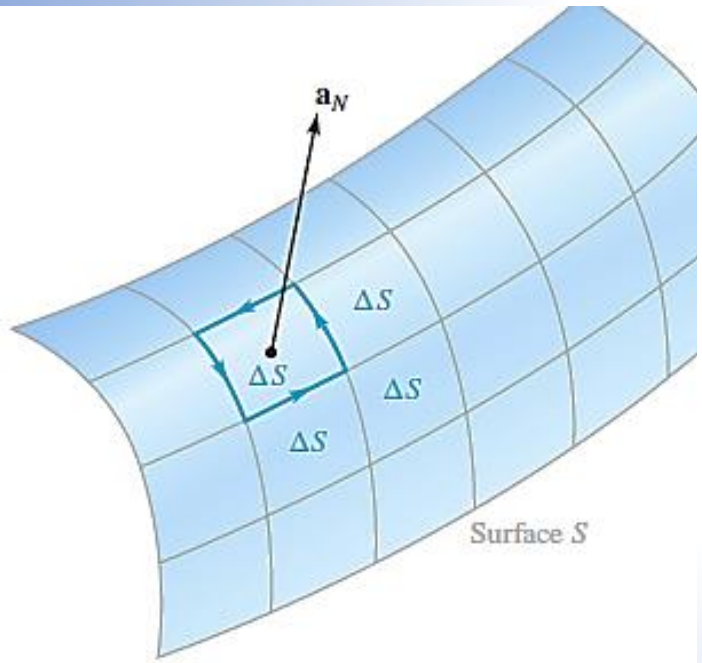


Figure 7.16 The sum of the closed line integrals about the perimeter of every ΔS is the same as the closed line integral about the perimeter of S because of cancellation on every interior path.

Proof: If we apply the definition of the curl to one of these incremental surfaces, then

$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}{\Delta S} \doteq (\nabla \times \mathbf{H})_N$$

$$\frac{\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S}}{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N$$

$$\oint \mathbf{H} \cdot d\mathbf{L}_{\Delta S} \doteq (\nabla \times \mathbf{H}) \cdot \mathbf{a}_N \Delta S = (\nabla \times \mathbf{H}) \cdot \Delta \mathbf{S}$$

Every *interior* wall is covered once in each direction.
The only boundaries on which cancellation cannot occur form the outside boundary, the path enclosing *S*.
Therefore we have

$$\oint \mathbf{H} \cdot d\mathbf{L} \equiv \int_S (\nabla \times \mathbf{H}) \cdot d\mathbf{S}$$

In cylindrical coordinates, the magnetic field is given as $\mathbf{H} = [4\rho - 2\rho^2]\,\mathbf{a}_\phi$ (A/m), $0 \le \rho \le 1$.

a. Find the current density as a function of ρ within the cylinder.

b. Find the total current that passes through the surface $z = 0$ and $0 \le \rho \le 1$ (m) in the $\mathbf{a}_z$ direction.

c. Check the result of part (b) by another method. — Stoke's theorem

$$\mathbf{J} = \nabla \times \mathbf{H}$$

$$= \frac{1}{\rho}\begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial\rho} & \frac{\partial}{\partial\phi} & \frac{\partial}{\partial z} \\ H_\rho & \rho H_\phi & H_z \end{vmatrix}$$

$$= \frac{1}{\rho}\begin{vmatrix} \mathbf{a}_\rho & \rho\mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial\rho} & 0 & 0 \\ 0 & \rho H_\phi & 0 \end{vmatrix}$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}(\rho\, H_\phi)\,\mathbf{a}_z$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}\left[\rho \times 4\rho - \rho \times 2\rho^2\right]\mathbf{a}_z$$

$$= \frac{1}{\rho}\frac{\partial}{\partial\rho}\left[4\rho^2 - 2\rho^3\right]\mathbf{a}_z$$

$$= \frac{1}{\rho}\left\{8\rho - 6\rho^2\right\}\mathbf{a}_z$$

$$= (8 - 6\rho)\,\mathbf{a}_z \ (\mathrm{A/m^2})$$

$$I = \int_S \mathbf{J}\cdot d\mathbf{S}$$

$$= \int_S J_z\,\mathbf{a}_z \cdot \rho\, d\rho\, d\phi\, \mathbf{a}_z$$

$$= \int_{\rho=0}^{1}\int_{\phi=0}^{2\pi} (8 - 6\rho)\rho\, d\rho\, d\phi$$

$$= 2\pi \int_{\rho=0}^{1} (8\rho - 6\rho^2)\, d\rho$$

$$= 2\pi \left\{\frac{8\rho^2}{2} - 6 \times \frac{\rho^3}{3}\right\}_{\rho=0}^{1}$$

$$= 2\pi \left\{4\rho^2 - 2\rho^3\right\}_0^1$$

$$= 2\pi \times 2 = 4\pi \ (\mathrm{A})$$

c) $I = \oint H \cdot dl$

$$= \int_{\phi=0}^{2\pi} (4\rho - 2\rho^2)\hat{a}_\phi \cdot \rho\, d\phi\, \hat{a}_\phi \Big]_{\rho=1}$$

$$= \int_0^{2\pi} (4-2)\cdot 1\, d\phi$$

$$= 2 \cdot 2\pi = 4\pi \ \mathrm{A}$$


Thank
you